29.04.2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: Jan 2025– April 2025		
Maximum Marks: 50 Examination: End-Semester Examination		Duration: 2 Hrs.
Programme code: 01/04 Programme: B. Tech in Computer Engineering/ B. Tech in Information Technology	Class: SY	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: Computer Engineering /Information Technology	
Course Code: 216U01C401 / 216U04C401	Name of the Course: Probability, Statistics and Optimization Technique	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	
i)	Calculate Decile D_4 and Percentile P_{80} for the data related to the age in years of 99 members in a housing society. Age (Years): 10, 18, 25, 35, 40, 45 No of Members: 20, 5, 10, 30, 20, 14	05
ii)	Find all basic solutions to the problem given below. Which of them are basic feasible, non-degenerate, and optimal basic feasible solutions? Maximize $z = 2x_1 + 4x_2 + x_3$ Subject to $x_1 + 3x_2 + 2x_3 = 5$, $2x_1 + x_2 + 4x_3 = 6$	05
iii)	A technical support hotline has a single customer service representative answering calls. Calls arrive randomly at an average rate of 10 calls per hour. Each call is handled independently, and the service time follows an exponential distribution with an average rate of 12 calls per hour. Find: 1. The probability that a caller has to wait before being served. 2. The expected number of callers in the queue. 3. The average time a caller spends in the system.	05
Q.2	Attempt any one	10
i)	(a) A researcher wants to determine whether gender and preference for a new product are independent. She surveys 200 people and records their responses as follows: Preference: Male, Female Likes: 50, 70 Dislikes: 30, 50 Using Chi Square test, check the independence at 1% LOS. (b) A university administrator claims that the average study time of students at the university is 25 hours per week. A researcher doubts this claim and conducts a survey of 100 students, recording their weekly study hours. The mean of the sample is 23.5 hours and standard deviation is 4.5 hours. Test whether the actual mean study time is different from 25 hours at a 5% significance level.	
ii)	Solve the following LPP using Simplex method Maximize $z = 5x_1 + 4x_2 + 3x_3$ Sub to $2x_1 + 3x_2 + x_3 \le 8$, $4x_1 + x_2 + 2x_3 \le 10$, $x_1 + 2x_2 + 3x_3 \le 6$, $x_1, x_2, x_3 \ge 0$	

Q.3	Attempt any one	10

i) A fitness trainer wants to determine whether a new diet plan significantly affects body weight. He collects data from 10 individuals by recording their weight before and after following the diet for 6 weeks.

Before Weight	80	75	90	85	88	76	92	81	77	84
After Weight	78	72	85	82	86	74	88	79	75	80

Test at 5% level of significance whether the diet significantly affects body weight.

ii) **(a)** The joint probability distribution of Y_1 and Y_2 is given by

$P(Y_1 = y_1, Y_2 = y_2) = \frac{1}{36}(y_1 + 3y_2)$ for $y_1 = 0,1,2,3$ and $y_2 = 0,1,2$

Find the joint probability mass function (p.m.f) of Y_1 and Y_2 and also determine the marginal probability distributions of Y_1 and Y_2.

(b) Determine the Karl Pearson's Coefficient of correlation between the number of hours studied (**X**) and the exam scores (**Y**) of 8 students.

Study Hour	2	4	6	8	10	12	14	16
Exam Score	50	55	65	70	75	80	85	90

Q.4	Attempt the following	10

(a) The lifespan (in hours) of a certain type of light bulb follows an exponential distribution with an average lifespan of 1000 hours. Find The probability that a randomly chosen light bulb lasts more than 1500 hours.Also find the probability that a light bulb fails within the first 500 hours.

(b) The regression lines of a sample are $2x + 5y = 20$, and $4x + 3y = 18$.

Find (i) Sample means $\bar{x}$ and $\bar{y}$,

(ii) coefficient of correlation between x and y.

(iii) Estimate y when $x = 12$.

Q.5	Attempt any two	
i)	The lifespans of a particular brand of fans are normally distributed with a mean of 1200 hours and a standard deviation of 200 hours. Find the percentage of fans having lifespan more than 1300 hours. If 1000 fans are bought by a consumer, how many will have lifespan less than 1150 hours?	05
ii)	Using the method of Lagrange's multipliers, solve the following NLPP Optimize $z = 4x_1^2 + 3x_2^2$ Subject to $x_1 + 4x_2 = 10$, $x_1, x_2 \geq 0$	05
iii)	A toll booth processes vehicles at a rate of 10 vehicles per hour (exponentially distributed). Vehicles arrive at the booth according to a Poisson process at a rate of 8 vehicles per hour. However, due to space limitations, only 3 vehicles can wait in the queue. If a vehicle arrives when the queue is full, it must leave. Find: 1. The probability that an arriving vehicle is denied service. 2. The average number of vehicles in the system. 3. The probability that the system is empty.	05